UNC GILLINGS SCHOOL OF GLOBAL PUBLIC HEALTH

Identification of key risk factors associated with severe hypoglycemia in older adults with Type 1 Diabetes

Nikki L. B. Freeman

13 December 2022

Outline

Objectives

Data

Methods
- Statistical approach
- Analysis framework
- Computation and data management

Results
- Predictive performance
- Identification of key predictors

Discussion

Acknowledgements

Setting and motivation

Our goal

- Extend the work of Weinstock et al. (2016)
- Use machine learning to model the relationship between severe hypoglycemia (SH) and associated risk factors.
- Use a data-driven strategy to identify a parsimonious subset of risk factors associated with SH.

Diabetes Care Volume 39, April 2016 603

Risk Factors Associated With Severe Hypoglycemia in Older Adults With Type 1 Diabetes

Diabetes Care 2016;39:603–610 | DOI: 10.2337/dc15-1426

*Ruth S. Weinstock,[1] Stephanie N. DuBose,[2] Richard M. Bergenstal,[3] Naomi S. Chaytor,[4] Christina Peterson,[5] Beth A. Olson,[1] Medha N. Munshi,[6] Alysa J.S. Perrin,[2] Kellee M. Miller,[2] Roy W. Beck,[2] David R. Liljenquist,[6] Grazia Aleppo,[7] John B. Buse,[8] Davida Kruger,[9] Anuj Bhargava,[10] Robin S. Goland,[11] Rachel C. Edelen,[12] Richard E. Pratley,[13] Anne L. Peters,[14] Henry Rodriguez,[15] Andrew J. Ahmann,[16] John-Paul Lock,[17] Satish K. Garg,[18] Michael R. Rickels,[19] and Irl B. Hirsch,[4] for the T1D Exchange Severe Hypoglycemia in Older Adults With Type 1 Diabetes Study Group**

Data

- Case-control study
  - 18 diabetes centers participating in the T1D Exchange Clinic Network
  - **Cases**: Had at least one SH event in the prior 12 months
  - **Control**: No SH event in the past 3 years
- **Severe hypoglycemia**: "An event requiring assistance of another person as a result of altered consciousness or confusion, to administer carbohydrate or glucagon or other resuscitative actions".
- **Inclusion**: Clinical diagnosis of autoimmune T1D being treated with insulin, age ≥ 60 years, and diabetes duration ≥ 20 years.
- **Exclusion**: Use of CGM, CKD stage 4 or 5, diagnosis of moderate or advanced dementia, serious illness with life expectancy of < 1 year, and history of pancreatic transplant.
- Measures: Demographics, medical history, diabetes management, physical exam, cognitive and functional tests

Statistical approach: modeling

We used 3 machine learning models

- Random forests (RF)
- $L1$-regularized logistic regression (LR)
- Support vector machines (SVM)

Assessment of predictive performance (all models)

- Accuracy, sensitivity, and specificity

Parsimonious lists of key predictors (RF and LR)

- Mean decrease in Gini-index (RF)
- Absolute value of the coefficients from the tuned model (LR)

Missing data and analysis framework

Analyses were conducted in an impute-train-test framework.

1. Multiple imputation on the full observed data set
2. 40% of the initially complete cases were randomly selected as the test set
3. Remaining cases and their imputed data sets were used as training sets
4. Models were trained on the training sets and then tested on the testing set; results were averaged across the training folds

Computation and data management

- Analyses were conducted in R
  - tidyverse, mice, caret, randomForest, glmnet, SVM, e1071

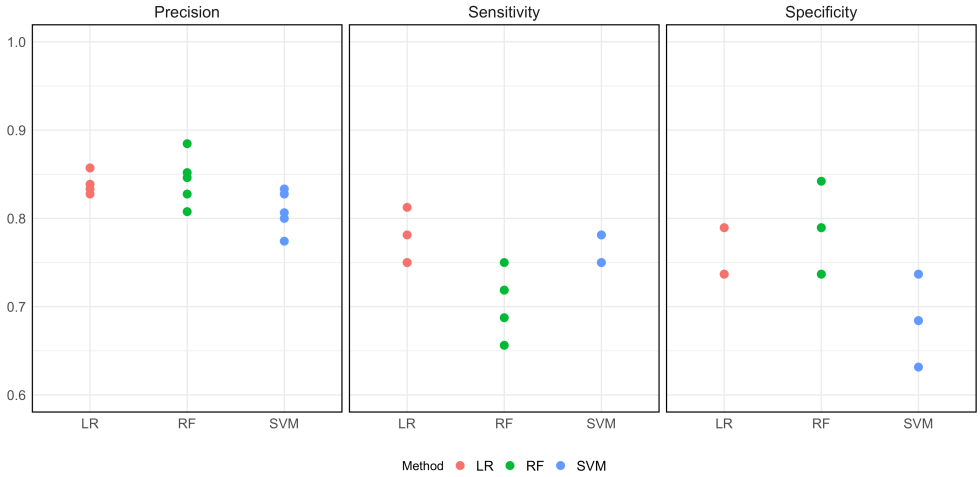
- The data and code for these analyses are publicly available at `https://github.com/nikkifreeman/T1D_SH_key_predictors`.

Participants and their characteristics

- The data set contained data for 203 study participants.
- Two participants were excluded from the analysis because they lacked any demographic data.
- The total study size was 201 participants with 101 cases and 100 controls.

	Case	Control	Overall
	(N=101)	(N=100)	(N=201)
Gender			
Female	51 (50.5%)	44 (44.0%)	95 (47.3%)
Male	50 (49.5%)	56 (56.0%)	106 (52.7%)
Race			
White	96 (95.0%)	95 (95.0%)	191 (95.0%)
Asian	0 (0%)	1 (1.0%)	1 (0.5%)
Unknown/not reported	1 (1.0%)	0 (0%)	1 (0.5%)
Black/African American	4 (4.0%)	3 (3.0%)	7 (3.5%)
American Indian/Alaskan Native	0 (0%)	1 (1.0%)	1 (0.5%)
Ethnicity			
Not Hispanic or Latino	98 (97.0%)	96 (96.0%)	194 (96.5%)
Unknown/not reported	0 (0%)	1 (1.0%)	1 (0.5%)
Hispanic or Latino	3 (3.0%)	3 (3.0%)	6 (3.0%)
Education			
<= HS	14 (13.9%)	11 (11.0%)	25 (12.4%)
Any college	59 (58.4%)	65 (65.0%)	124 (61.7%)
Advanced degree	27 (26.7%)	23 (23.0%)	50 (24.9%)
Missing	1 (1.0%)	1 (1.0%)	2 (1.0%)
Any insurance			
No	1 (1.0%)	2 (2.0%)	3 (1.5%)
Yes	99 (98.0%)	98 (98.0%)	197 (98.0%)
Missing	1 (1.0%)	0 (0%)	1 (0.5%)
BMI category			
Underweight	3 (3.0%)	0 (0%)	3 (1.5%)
Normal weight	35 (34.7%)	36 (36.0%)	71 (35.3%)
Overweight	36 (35.6%)	37 (37.0%)	73 (36.3%)
Obese	25 (24.8%)	25 (25.0%)	50 (24.9%)
Missing	2 (2.0%)	2 (2.0%)	4 (2.0%)

Predictive performance



Predictive performance

	LR	RF	SVM
Accuracy	0.76	0.73	0.74
Precision	0.84	0.84	0.81
Sensitivity	0.77	0.71	0.76
Specificity	0.76	0.78	0.69

Key predictors

Random Forest

Fold 1	Fold 2
Hypoglycemia unawareness score	Hypoglycemia unawareness score
Symbolic digit modalities (written)	Symbolic digit modalities (written)
Grooved peg board (dominant hand)	HBA1C
Trail making test	Trail making test
HBA1C	Units basal or long acting insulin
Fold 3	**Fold 4**
Hypoglycemia unawareness score	Hypoglycemia unawareness score
Symbolic digit modalities (written)	Symbolic digit modalities (written)
Trailing making test	HBA1C
HBA1C	Grooved peg board (dominant hand)
Grooved peg board (dominant hand)	Trail making test
Fold 5	
Hypoglycemia unawareness score	
Symbolic digit modalities (written)	
Trail making test	
Grooved peg board (dominant hand)	
Units basal or long acting insulin	

$L1$-regularized Logistic Regression

Fold 1	Fold 2
Hypoglycemia unawareness	Hypoglycemia reduced awareness
Underweight	Hypoglycemia unawareness
Hypoglycemia reduced awareness	Underweight
Hispanic	Beta blockers
Beta blockers	Hispanic
Fold 3	**Fold 4**
Hypoglycemia unawareness	Hypoglycemia unawareness
Hypoglycemia reduced awareness	Hypoglycemia reduced awareness
Underweight	Hispanic
Hispanic	Underweight
Beta blockers	Beta blockers
Fold 5	
Hypoglycemia unawareness	
Hypoglycemia reduced awareness	
Underweight	
Hispanic	
Beta blockers	

Discussion

- Our trained models performed well with accuracy above 70% for all models, and hypoglycemia unawareness was the top predictor for both the RF and LR models.

- Limitations
  - Modest study size
  - Generalizability

- Conclusions and recommendations
  - Our work demonstrates the feasibility of using machine learning to generate data-driven prediction models of SH in older adults with T1D
  - Evidence for further exploration of interventional strategies to increase hypoglycemia awareness
  - Future work should focus on building more diverse data sets to improve the generalizability of prediction models

Thank you!

- **Dr. M. Victor Wickerhauser**, project advisor
- **Dr. Lisa Kuehne**, second reader
- **Dr. Anna Kahkoska**, T1D expert
- **Elizabeth Fogt**, advisor